

Python Lists: Shorthand Study Guide

1. Basics & Properties

- **Definition:** Lists are used to store multiple items in a single variable. They are created using square brackets: `thislist = ["apple", "banana", "cherry"]`.
- **Ordered:** Items have a defined order that will not change. New items are added to the end.
- **Changeable:** You can change, add, and remove items after the list is created.
- **Allow Duplicates:** Because they are indexed, lists can have multiple items with the same value.
- **Data Types:** Lists can store any data type (strings, integers, booleans) and even mixed types in one list.
- **Constructor:** You can also use the `list()` constructor to create a list.

2. Accessing & Modifying Items

- **Access:** Use index numbers in square brackets. ****Index**** is the first item.
 - *Example:* `print(mylist)` outputs the second item.
- **Length:** Use `len(thislist)` to find the number of items.

3. Adding & Removing (Shorthand)

- **Add Items:**
 - `append()`: Adds an item to the end of the list.
 - `insert(index, val)`: Adds an item at a specific position.
 - `extend()`: Appends elements from another list (or any iterable) to the current list.
- **Remove Items:**
 - `remove(val)`: Removes the first occurrence of a specific value.
 - `pop(index)`: Removes the item at the given index (or the last item if no index is specified).
 - `clear()`: Empties the entire list.

4. Looping & Comprehension

- **Looping:** Use a for loop to iterate through items: `for x in thislist: print(x)`.
- **List Comprehension:** A shorter syntax for creating a new list based on the values of an existing list.
 - *Example:* `newlist = [x for x in fruits if "a" in x]`

5. Sorting, Copying, & Joining

- **Sort:** Use `.sort()` to sort the list alphanumerically by default.
- **Copy:** Use `.copy()` or the `list()` constructor to make a copy of a list, as simply assigning `list2 = list1` only creates a reference.
- **Join:** Use the `+` operator or the `.extend()` method to combine two lists.

6. Essential List Methods Reference

- `append()`, `clear()`, `copy()`, `count()`, `extend()`, `index()`, `insert()`, `pop()`, `remove()`, `reverse()`, `sort()`.